

[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Monday, 24 August 2026, 8:12 PM**State** Finished**Completed on** Monday, 24 August 2026, 8:23 PM**Time taken** 10 mins 48 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZero(int a[], int l, int h)
4  {
5      if (l > h)
6          return 0;
7
8      if (a[l] == 0)
9          return h - l + 1;
10
11     int m = (l + h) / 2;
12
13     return countZero(a, m + 1, h);
14 }
15
16 int main()
17 {
18     int n, i;
19     scanf("%d", &n);
20
21     int a[n];
22
23     for (i = 0; i < n; i++)
24         scanf("%d", &a[i]);
25
26     printf("%d", countZero(a, 0, n - 1));
27
28     return 0;
29 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 27 August 2026, 11:12 AM**State** Finished**Completed on** Thursday, 27 August 2026, 11:23 AM**Time taken** 11 mins 30 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     int m=n/2;
6     int arr[n];
7     int freq[1000]={0};
8     for(int i=0;i<n;i++){
9         scanf("%d",&arr[i]);
10        freq[arr[i]]++;
11    }
12    for(int i=0;i<n;i++){
13        if(freq[arr[i]]>m){
14            printf("%d",arr[i]);
15            break;
16        }
17    }
18 }
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 27 August 2026, 11:33 AM**State** Finished**Completed on** Monday, 31 August 2026, 10:00 PM**Time taken** 4 days 10 hours**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, x;
5      scanf("%d", &n);
6      int a[n];
7      for (int i = 0; i < n; i++) scanf("%d", &a[i]);
8      scanf("%d", &x);
9
10     int l = 0, h = n - 1, ans = -1;
11     while (l <= h) {
12         int m = (l + h) / 2;
13         if (a[m] <= x) {
14             ans = a[m];
15             l = m + 1;
16         } else {
17             h = m - 1;
18         }
19     }
20     printf("%d\n", ans);
21     return 0;
22 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓

	Input	Expected	Got	
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on Monday, 31 August 2026, 10:03 PM

State Finished

Completed on Monday, 31 August 2026, 10:04 PM

Time taken 1 min 13 secs

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main()
3  {
4      int n,x;
5      scanf("%d",&n);
6      int arr[n];
7      for(int i=0;i<n;i++)
8          scanf("%d",&arr[i]);
9      scanf("%d",&x);
10     for(int i=0;i<n;i++)
11     {
12         int target=x-arr[i];
13         int left=i+1,right=n-1;
14         while(left<=right)
15         {
16             int mid=(left+right)/2;
17             if(arr[mid]==target)
18             {
19                 printf("%d\n",arr[i]);
20                 printf("%d\n",arr[mid]);
21                 return 0;
22             }
23             else if(arr[mid]<target)
24                 left=mid+1;
25             else
26                 right=mid-1;
27         }
28     }
29     printf("No\n");
30     return 0;
31 }
```

	Input	Expected	Got	
✓	4 2 4 8 10 14	4 10	4 10	✓
✓	5 2 4 6 8 10 100	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Monday, 31 August 2026, 10:05 PM**State** Finished**Completed on** Monday, 31 August 2026, 10:06 PM**Time taken** 29 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

1  #include <stdio.h>
2  void quicksort(int a[],int low,int high)
3  {
4      if(low<high)
5      {
6          int pivot=a[high];
7          int i=low-1;
8          for(int j=low;j<high;j++)
9          {
10             if(a[j]<pivot)
11             {
12                 i++;
13                 int t=a[i];
14                 a[i]=a[j];
15                 a[j]=t;
16             }
17         }
18         int t=a[i+1];
19         a[i+1]=a[high];
20         a[high]=t;
21         int p=i+1;
22         quicksort(a,low,p-1);
23         quicksort(a,p+1,high);
24     }
25 }
26 int main()
27 {
28     int n;
29     scanf("%d",&n);
30     int a[n];
31     for(int i=0;i<n;i++)
32         scanf("%d",&a[i]);
33     quicksort(a,0,n-1);
34     for(int i=0;i<n;i++)
35         printf("%d ",a[i]);
36     return 0;
37 }
```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓

	Input	Expected	Got	
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)